

VIRTUAL 3D TOUR ASSISTANCE IN REAL ESTATE MANAGEMENT

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Abstract. Over the past decades, more efficient real estate management opportunities have significantly expanded and improved. Complex methods, innovative techniques, and a constantly evolving spectrum of information technology have become readily available for common use. When managing real estate for short-term or medium-term rentals, planning, servicing, and ensuring accommodation quality are inseparable aspects, as is providing suitable housing. This process begins at the very first step when potential tenants request a viewing of the preferred property. The situation of the past few years demonstrated that the real estate business was one of the most affected economic sectors by the COVID-19 pandemic. Potential buyers or tenants could no longer physically visit and experience the space as before. Faced with this new reality, there was a need to find a tool to enhance the real estate management strategy by presenting the property. One such tool became the 360-degree photography and virtual tour. The aim of this study is to assess the practical applicability of 3D video tours and evaluate the advantages and drawbacks in short- and medium-term rental business.

Keywords: *Real estate management, 3D video tours, innovative techniques, short- and medium-term rental business.*

INTRODUCTION

Due to technological advancements (and a global pandemic), virtual tours in the real estate industry are becoming increasingly popular. According to Forbes (2022), 61 % of homebuyers considered virtual tours as the most useful technology for viewing properties, surpassing traditional options such as high-quality photographs (51 %) and accurate information presentation (58 %). Virtual tours have gained significant importance in the real estate industry, as they provide a unique opportunity for potential buyers to explore properties online and allow agents to showcase real estate to a broader audience. This not only saves agents' time in organizing viewings but also reduces expenses by decreasing the number of physical property visits in different locations.

The 3D tour technology creates the most accurate and immersive virtual home-buying experience in the market. By capturing each space in a unique format, a single digital twin can generate floor plans, 3D virtual tours, and any number of HDR photos for each room, from any angle, at any time. By utilizing this innovative

technology, any real estate professional can leverage the power of digital twins in their marketing toolkit, making it more effective and easier to sell more properties to highly interested buyers (Matterport, 2023a).

Matterport is a leading spatial data company in the United States, focused on creating and indexing the digital world. Their all-in-one 3D data platform transforms spaces into precise and engaging digital twins, which can be used in design, construction, operation, advertising, and understanding any space. This internet-based 3D virtual tour technology converts 360-degree photos into interactive 3D representations of property space. It offers an innovative, comprehensive, and immersive media platform that allows creators to capture and share content 24/7. Any potential buyer can be transported to the space at any time and from anywhere without any expenses or commitments. Moreover, Matterport is easily accessible to users, as it is optimized for mobile devices (Fig. 1). The program provides a “do-it-yourself” feature, enabling creators to capture 360-degree photos using their smartphones and then upload them to the Matterport web platform. The remaining processing is done by Matterport’s scanning software, which aligns, recalibrates, and stitches each frame together, turning them into a 3D model space. Matterport also allows users to share their 3D models anywhere and at any time (Matterport, 2023b).



Fig. 1. 3D tour availability via mobile tour.

1. IMPLEMENTATION OF 3D VIRTUAL TOURS

Real estate companies utilize various platforms for property rentals. Short-term rental activities are carried out through major platforms like Airbnb.com and Booking.com. After conducting a search for innovative and technology-oriented companies in Lithuania, the company “1001 Pikselis” was found, which provides

services to assist in real estate management. A 3D tour scanning was performed at the property of the real estate management company "Domillion" located in Vilnius.

1.1. 3D virtual tour scanning

Using unique technologies and 3D cameras, a 99 % accurate 3D model of the premises is created, allowing virtual walkthroughs from any location in the world at any time. It serves as a digital twin of the space, available online. The 3D tour is accessible through a link, eliminating the need to upload additional files to your server. A short HTML code provided by the service provider seamlessly integrates the 3D tour into your existing website, ensuring that the client does not get redirected to an external link. And all of this can be done in less than 48 hours (1001pikselis, 2023).

The 3D model of the space allows you to view the premises from various angles from any perspective. There is an option to view each floor separately or the entire model at once. This provides the flexibility to explore the property from different viewpoints, offering a comprehensive understanding of the space's layout and design (Fig. 2) (Sulaiman, M., et al., 2020).

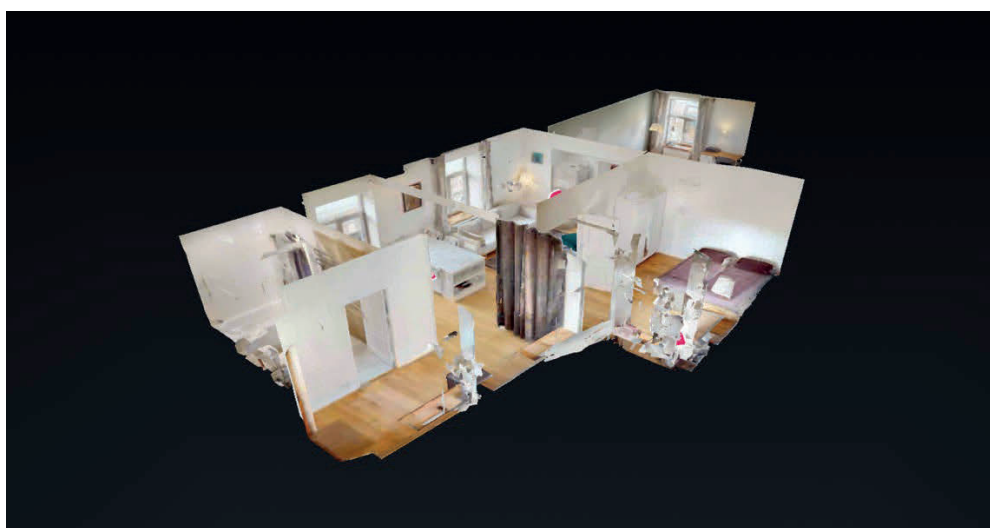


Fig. 2. 3D apartment layout: top view of the space.

1.2. 3D tour management

The 3D virtual tour allows independent exploration of the premises, enabling potential tenants to walk through the rooms and view everything from the layout to the design, emotionally engaging viewers from the very beginning. The visualization platform utilizes patented software to create 360-degree interactive images (Fig. 3). There is no need to install specific software or technical equipment to create virtual real estate tours; the tours are demonstrated through a single saved link.

By using virtual apartment viewings, clients can review properties as many times as they wish, saving time and fuel costs traveling from one location to another. This makes it easier for them to compare different real estate properties and decide

which one they want to buy or rent. Additionally, if the buyer's family lives in different regions, all members can still participate without being physically present. This can sometimes be a crucial factor when investing in real estate.

Each 3D tour comes with an integrated measurement feature that allows easy measurement of the dimensions of the relevant spaces. This is particularly useful for tenants who are looking to integrate their existing furniture into the apartment and need measurements for doors or specific areas of a room (Fig. 4). For real estate managers, this feature saves time by eliminating the need to physically measure spaces based on specific client requests while traveling to various properties.

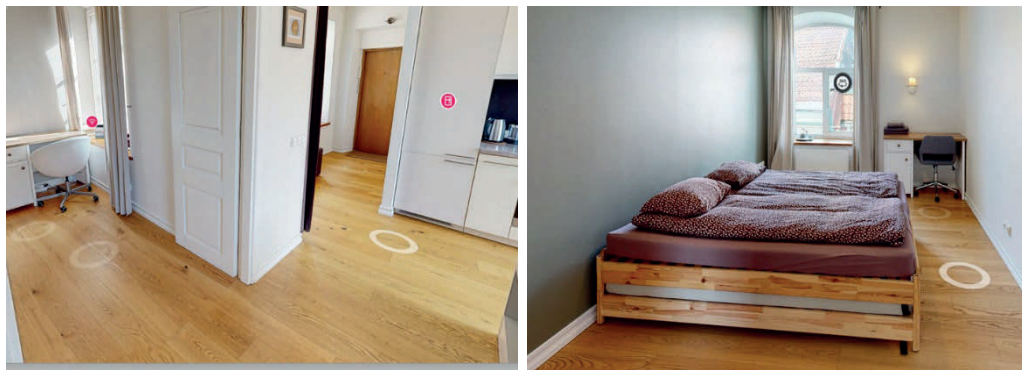


Fig. 3. 3D virtual tour of the short-term-rental apartment.

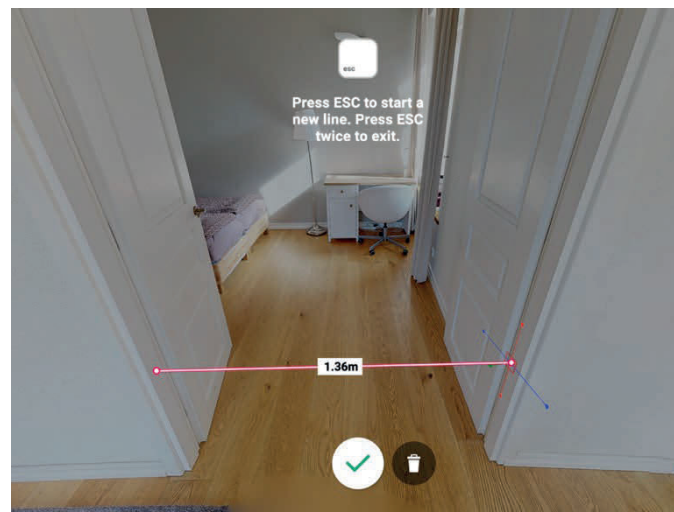


Fig. 4. Measuring feature.

The small circles, referred to as information points, possess great power by emphasizing the unique places and features of the rooms, revealing integrated objects that may not be visible in the photos (Fig. 5). These information points can contain textual and visual information, documents, or links to web pages. For tenants, the use of information points is relevant during property inspections and when moving in, as integrated objects may have attached instructions or video

materials that allow tenants to familiarize themselves with the devices without seeking assistance from the owners. On the other hand, for managers, information points help improve efficiency by saving time when addressing the same recurring questions from tenants.

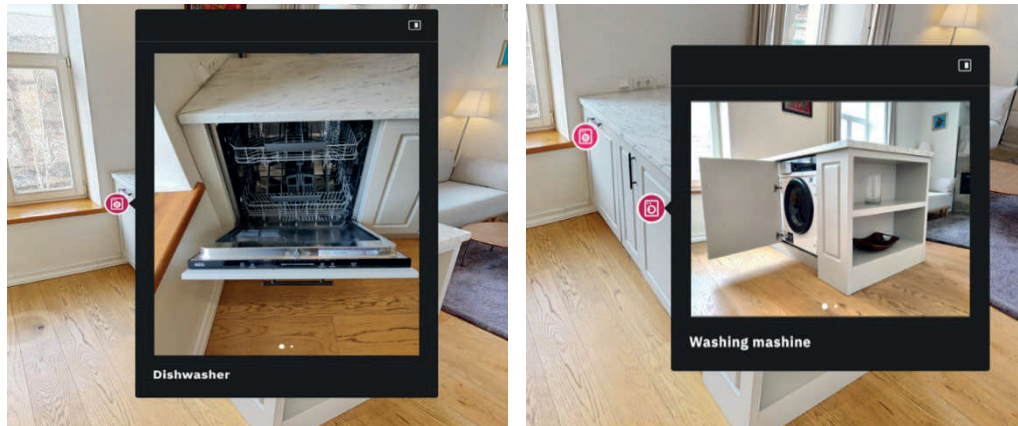


Fig. 5. Information points.

Virtual 3D tours can be exported to Google Street View, making you even more visible and allowing clients to easily orient themselves regarding the location of the apartment within the city. This feature is also useful for evaluating the accessibility by car or analyzing the intensity of the street traffic. Moreover, it enables clients to assess the direction which the apartment's windows face and the views visible through them. The integration with Google Street View enhances the overall experience and provides valuable information for potential buyers or tenants, making the decision-making process more informed and convenient.

This testing innovation comes to a common understanding that it serves as an educational employee tool:

1. It simplifies and facilitates the introduction of employees to managed properties. Employees in customer service roles need to be well acquainted with all managed properties to answer any customer questions. With the help of the virtual tour, they can virtually enter an apartment and look around to see what items are present, the orientation of windows, the type and size of beds, and more. This eliminates the need to ask other colleagues, physically visit a specific property, or leave customer inquiries unanswered.
2. It is a training tool for property managers. As this group of employees experiences the most turnover, training on how to organize an apartment, set up beds, or place different types of furniture can be simulated through this virtual tour. Additionally, after a longer-term, items in the apartment are often moved around by tenants. By clicking on the link and virtually entering the apartment, property managers can see where the furniture had been originally placed.
3. It is of assistance for maintenance workers. When a device malfunctions and requires parts replacement, it is often essential to know the device's manufacturer. The virtual tour is precise enough for property managers to

view the manufacturer's logo on the device, ensuring an efficient problem-solving process.

1.3. Data Analysis

Undoubtedly, this technology brings numerous advantages and proves to be highly effective in real estate management. Practical observation revealed strengths and weaknesses of 3D for short- and medium-term rental business.

Table 1. Strengths and Weaknesses of 3D Tour for Short- and Medium-term Rental Business

Strengths	Weaknesses
- Better perception of the apartment's layout and size for tenants - More engaging impressions for potential clients - Time-saving for both tenants and property managers - Competitive advantage through innovative implementation in the company - Interactive experience - Increased work efficiency by saving employees' time	- Unpredictable return on investment for implementing video tours - Ongoing data maintenance costs - Expenses for preparing the apartment for the tour - Often the visual appeal of staircases in old town buildings is limited - Lack of personal interaction with potential tenants - If the video tour experiences technical issues or malfunctions for any reason, potential tenants may become disappointed and lose interest in the property

3D tours are not inexpensive products, especially when the number of properties to be scanned is in the tens. In this case, the analyzed object on Žemaitijos g. has a size of 60 m² and the scanning cost is 120 euros, excluding taxes. The on-site visit with special equipment and the scanning of the spaces are one-time actions and are paid as a one-time fee. However, about 20 % of the total cost accounts for the annual fee paid to the service provider for data maintenance on their servers.

Information points are priced per the required number of units. One information point costs 3 euros without taxes. If the client wants to have the option of an automatic tour, such a feature is available but comes with an additional cost. In the case of the analyzed property, the auto-tour service was not ordered. An estimate for the service cost that would be incurred by a real estate management company after performing 50 apartment scans would cost approximately 10 000 euros.

CONCLUSIONS

The adaptability of 3D virtual tours for real estate inspections has provided the opportunity not only to save on inspection travel expenses but also to save time for both tenants and landlords, while also providing a physical sense of the space. Such virtual tours can change the working habits of real estate managers and offer clients more engaging and attractive ways of interacting with the surrounding world. Incorporating interactive 3D tours into your short-term rental business can lead to increased inquiries, higher quality leads, and ultimately, more bookings. It is a valuable tool for enhancing the guest experience, streamlining the rental process, and differentiating your properties in a competitive market. For the company embracing technology trends shows that the business is forward-thinking and in tune with modern advancements, making short-term rental business more appealing to tech-savvy renters.

Despite the numerous advantages, 3D virtual tours can be expensive for a large number of real estate properties. The investments can amount to tens of thousands of euros, and the return on investment can be challenging, especially for businesses focused on short-term rentals. Online platforms for short-term rentals often do not allow the use of foreign products, limiting clients to photos only. After conducting an internal analysis, it is concluded that the most effective way to utilize this technology would be in single buildings or business centers. This would offer the opportunity to comprehensively cover the entire property.

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